

**AN OCR-BASED REAL ESTATE MANAGEMENT SYSTEM
FOR ST. FRANCIS SQUARE REALTY CORPORATION
USING LEVENSHTein DISTANCE FOR
AUTOMATED DOCUMENT
VALIDATION**

In Partial Fulfillment of the Requirements for the Degree of
BACHELOR OF SCIENCE IN COMPUTER SCIENCE

by

<Month and Year of Graduation>

ABSTRACT

Real estate companies handle numerous business processes involving property management, client transactions, billing, payment monitoring, and document verification. At St. Francis Square Realty Corp., many of these processes are performed manually, resulting in time-consuming document validation, fragmented record management, delayed transaction processing, and an increased risk of human error. To address these challenges, this study proposes the development of an **OCR-Based Real Estate Management System for St. Francis Square Realty Corp. Using Levenshtein Distance for Automated Document Validation**.

The primary objective of this study is to design and develop two integrated systems: an **Internal Management System** for authorized company personnel and a **Web-Based Client Portal** for buyers. The Internal Management System will facilitate property management, client records, billing, payment monitoring, document management, report generation, and user administration. The Web-Based Client Portal will enable clients to register, browse available properties, submit reservation requests, upload documentary requirements and proof of payment, monitor transaction status, and receive system notifications.

To improve document validation, the proposed system will integrate Optical Character Recognition (OCR) technology to extract text from uploaded documents. The extracted information will then be compared with the company's database using the Levenshtein Distance algorithm, allowing the system to identify minor OCR recognition errors while accurately validating client information. Documents that satisfy the required similarity threshold will be accepted automatically, while those with significant discrepancies will be flagged for manual verification.

The study will employ the Agile Software Development Life Cycle (SDLC) methodology in designing, developing, implementing, and testing the proposed systems. The integrated solution is expected to improve operational efficiency, reduce manual document processing, enhance billing and payment monitoring, strengthen document validation, and provide clients with convenient online access to real estate services. Furthermore, the study is expected to serve as a reference for future researchers and organizations seeking to implement OCR-assisted document validation and web-based real estate management solutions.

Keywords: Optical Character Recognition (OCR), Real Estate Management System, Web-Based Client Portal, Internal Management System, Levenshtein Distance, Document Validation, Payment Monitoring

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CHAPTER I

PROJECT AND ITS BACKGROUND

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The continuous advancement of **Information and Communication Technology (ICT)** has significantly transformed the way organizations conduct business, manage information, and deliver services. The integration of digital technologies into organizational operations, commonly referred to as **digital transformation**, has enabled businesses to replace traditional manual processes with automated and technology-driven solutions that improve operational efficiency, service quality, and organizational productivity. As businesses continue to operate in an increasingly competitive and technology-oriented environment, digital transformation has become an essential strategy for improving operational performance, reducing processing time, supporting informed decision-making, and enhancing customer satisfaction (Laudon & Laudon, 2022).

One of the primary drivers of digital transformation is the implementation of **Information Systems (IS)**. An information system is an organized combination of people, hardware, software, databases, communication networks, and procedures that work together to collect, process, store, and distribute information to support organizational operations and decision-making (Laudon & Laudon, 2022). Information systems enable organizations to automate repetitive business activities, centralize

organizational records, improve communication among departments, and generate timely and accurate information. Through automation, organizations can minimize manual processing, reduce human error, increase productivity, and provide faster and more reliable services to customers.

Among the different types of information systems, **Web-Based Information Systems (WBIS)** have become one of the most widely adopted technologies across various industries. A web-based information system is an application that enables users to access information and perform transactions through a web browser using an Internet or local network connection. Unlike traditional desktop applications that are limited to specific computers, web-based systems operate using a **client-server architecture**, wherein user requests are processed by a centralized server connected to a centralized database. This architecture enables multiple users to access the system simultaneously while maintaining data consistency, integrity, and security. Furthermore, web-based systems support **Role-Based Access Control (RBAC)**, allowing users to access only the information and functions that correspond to their assigned responsibilities, thereby strengthening data security and accountability (Sandhu et al., 2000).

The growing adoption of web-based information systems reflects the continuing trend toward digitalization and online service delivery. Organizations are increasingly implementing customer self-service portals, electronic transactions, paperless documentation, and integrated information systems to improve operational efficiency and customer experience. These technologies have transformed the way organizations interact with their customers by allowing transactions to be completed remotely through Internet-enabled devices. As a result, businesses are able to provide faster services,

improve communication, reduce paperwork, and maintain centralized information that can be accessed securely by authorized users anytime and anywhere.

The implementation of web-based information systems offers numerous advantages to modern organizations. One of the most significant benefits is **centralized information management**, where organizational records are stored in a single database that serves as the authoritative source of information for all authorized users. This approach minimizes data redundancy, improves information consistency, and facilitates real-time information sharing among departments. Web-based systems also promote **remote accessibility**, allowing employees and customers to access services without geographical limitations. In addition, these systems provide **scalability**, enabling organizations to accommodate increasing numbers of users, transactions, and additional functionalities as business requirements evolve. Organizations likewise achieve **cost efficiency** by reducing paper consumption, minimizing repetitive administrative work, and simplifying software deployment and maintenance since updates are performed centrally rather than on individual workstations.

The widespread use of web-based information systems demonstrates their versatility across various industries. Financial institutions utilize these systems to provide online banking and electronic payment services. Healthcare organizations implement web-based systems for electronic medical records and appointment scheduling. Educational institutions rely on online enrollment systems and learning management platforms to improve academic administration. Government agencies have likewise adopted electronic government (e-government) services to facilitate online applications, licensing, and tax payments. Similarly, the **real estate industry** has

increasingly adopted web-based technologies to modernize property marketing, reservation management, billing, payment monitoring, and customer service.

The real estate industry manages large volumes of information related to property inventories, client records, reservations, contracts, documentary requirements, billing, payment monitoring, and after-sales services. These business processes require accurate information management and efficient coordination among different departments to ensure that transactions are completed accurately and promptly. Traditionally, many of these activities involve manual document submission, repetitive data encoding, physical record keeping, and manual verification of documentary requirements. Although these procedures remain functional, they often consume significant time and resources, increase administrative workload, and create opportunities for human error, particularly when handling numerous client transactions.

To improve operational efficiency, many real estate organizations have implemented **Real Estate Management Systems (REMS)** that integrate multiple business functions into a centralized platform. A Real Estate Management System automates property management, reservation processing, client information management, billing, payment monitoring, document management, report generation, and transaction tracking. By integrating these functions into a single information system, organizations are able to streamline workflows, improve information accuracy, strengthen communication among departments, and enhance customer service. Modern REMS also support customer-facing portals, enabling clients to perform selected transactions online while allowing organizations to efficiently manage internal business operations.

Today's real estate customers likewise expect convenient digital services throughout the property acquisition process. Rather than relying solely on face-to-face transactions, prospective buyers increasingly prefer online platforms where they can browse available properties, request property viewing schedules, reserve units, submit documentary requirements, upload proof of payment, and monitor the progress of their transactions. Consequently, many real estate companies have adopted **Web-Based Real Estate Portals** to provide clients with secure and convenient access to these services while simultaneously improving operational efficiency through automation and centralized information management.

This increasing reliance on digital transactions has also emphasized the importance of efficient **electronic document management**. Real estate transactions require clients to submit numerous documentary requirements, including government-issued identification cards, reservation forms, proof of payment, Special Power of Attorney (SPA), contracts, and other supporting documents. These documents contain essential information that must be verified before reservation approval, billing preparation, and other succeeding business processes can proceed. In many organizations, document verification continues to be performed manually, requiring employees to inspect uploaded files, compare information against existing records, and encode relevant data into the organization's database. While effective, this manual approach is time-consuming, repetitive, and susceptible to encoding errors, particularly when processing a large volume of client documents.

As organizations continue to digitalize their business operations, efficient document management has become an essential component of modern information

systems. Documents serve as the primary source of information for verifying customer identity, confirming transactions, ensuring regulatory compliance, and maintaining accurate organizational records. In the real estate industry, documentary requirements play a significant role throughout the property acquisition process, as clients are required to submit valid government-issued identification cards, reservation forms, proof of payment, contracts, Special Power of Attorney (SPA), and other supporting documents before transactions can proceed. The accuracy and timely verification of these documents are therefore critical to ensuring efficient business operations and protecting the integrity of organizational records.

Traditionally, document verification requires employees to manually inspect each uploaded document and compare the information against existing company records. This process involves reading document contents, encoding relevant information into the system, and verifying whether the submitted information matches the organization's database. Although manual verification helps ensure data accuracy, it is repetitive, labor-intensive, and prone to human error, particularly when handling large volumes of transactions. Minor encoding mistakes, overlooked information, and inconsistent validation procedures may result in delayed transaction processing and increased administrative workload.

To improve document processing efficiency, organizations have increasingly adopted **Optical Character Recognition (OCR)** technology. OCR is an intelligent document processing technology capable of converting printed or scanned text from images and Portable Document Format (PDF) files into machine-readable and editable text. Rather than requiring employees to manually encode information from every

uploaded document, OCR automatically extracts relevant information such as names, identification numbers, document numbers, dates, and other essential details. The use of OCR significantly reduces repetitive data encoding, accelerates document processing, improves operational efficiency, and minimizes human intervention during document verification.

Despite its advantages, OCR technology is not completely immune to recognition errors. The quality of the uploaded document greatly influences the accuracy of text extraction. Poor lighting conditions, blurred images, low scanning resolution, shadows, damaged documents, complex backgrounds, and font variations may cause OCR to incorrectly recognize certain characters or words. Consequently, even a single incorrectly recognized character may prevent an exact comparison from matching the extracted information with the corresponding record stored in the database. This limitation demonstrates that OCR alone is insufficient for achieving reliable automated document validation.

To overcome this limitation, intelligent text comparison algorithms are incorporated into modern document processing systems. Among these algorithms, the **Levenshtein Distance Algorithm** has been widely used because of its ability to measure the similarity between two text strings. Rather than requiring an exact match, the algorithm calculates the minimum number of character insertions, deletions, or substitutions necessary to transform one string into another. By determining the degree of similarity between two text values, the algorithm can recognize minor variations caused by OCR recognition errors while still identifying that the extracted information corresponds to the correct record. This capability improves document validation

accuracy and significantly reduces false mismatches that would otherwise require manual verification.

The practical application of these technologies is highly relevant to the operations of **St. Francis Square Realty Corp.**, a real estate company engaged in the marketing and sale of condominium units. Similar to many real estate organizations, the company manages numerous transactions involving property reservations, documentary requirements, client records, billing, payment monitoring, and after-sales services. These activities require accurate information management and close coordination among different departments to ensure that transactions are processed efficiently and that organizational records remain complete and reliable.

At present, several business processes require considerable manual intervention. Prospective buyers submit documentary requirements and proof of payment, which must be reviewed and verified by authorized personnel before reservation approval and other succeeding transactions can proceed. Employees manually compare information contained in uploaded documents with existing company records and encode the necessary information into the organization's database. Reservation monitoring, billing preparation, payment verification, and transaction tracking likewise require coordination among different personnel, making the overall process time-consuming and increasing the possibility of processing delays and human error as transaction volumes continue to grow.

Another identified limitation is the availability of online services for prospective buyers and existing clients. Although clients can obtain information regarding available properties, many succeeding transactions still require continuous communication with

company personnel through conventional methods. As a result, clients may experience delays in submitting documentary requirements, confirming payments, receiving transaction updates, and monitoring the progress of their reservations. These limitations highlight the need for a more integrated and customer-oriented information system capable of supporting both organizational operations and online client services.

In response to these operational challenges, this study proposes the development of an **OCR-Based Real Estate Management System for St. Francis Square Realty Corp. Using the Levenshtein Distance Algorithm for Automated Document Validation**. The proposed solution is an **Integrated Real Estate Management System** consisting of two interconnected platforms that share a centralized database.

The first platform is the **Web-Based Real Estate Portal**, which serves as the company's online service platform for prospective buyers and existing clients. Individuals may browse available properties, view project information, and explore unit details without creating an account. However, prospective buyers who intend to request a tripping schedule or reserve a unit are required to register and create an **Initial Account**. This account enables users to request tripping appointments, submit reservation requests, upload documentary requirements, upload proof of payment, receive transaction notifications, and monitor the progress of their reservation. Once the reservation has been reviewed and officially approved by the company, the Initial Account is converted into a **Permanent Client Account**, providing access to the **Client Portal**. Through the Client Portal, clients can conveniently monitor their Statement of Account, billing records, payment history, uploaded documents, transaction status,

announcements, and other after-sales services throughout the property acquisition and ownership process.

The second platform is the **Internal Management System**, which is exclusively accessible to authorized employees. The system implements **Role-Based Access Control (RBAC)** to ensure that users can only access the modules and functions corresponding to their assigned responsibilities. Depending on their assigned roles, authorized personnel may manage property inventories, reservation processing, client information, document verification, billing generation, payment monitoring, report generation, user administration, and transaction monitoring. Both systems operate using a centralized database, enabling real-time information sharing, improving coordination among departments, and maintaining consistent and accurate organizational records.

A significant innovation of the proposed system is the integration of **Optical Character Recognition (OCR)** and the **Levenshtein Distance Algorithm** for automated document validation. During document submission, OCR extracts relevant textual information from uploaded documents, while the Levenshtein Distance Algorithm compares the extracted information with records stored in the centralized database to determine their similarity. This approach enables the system to tolerate minor OCR recognition errors while maintaining accurate document validation, thereby reducing repetitive manual encoding, minimizing verification errors, accelerating document processing, and improving overall operational efficiency.

Ultimately, the proposed Integrated Real Estate Management System aims to modernize the existing business processes of St. Francis Square Realty Corp. by integrating online client services with internal business operations into a single

technology-driven platform. The system is expected to improve reservation management, strengthen document validation, streamline billing and payment monitoring, enhance customer experience, and support more efficient decision-making through centralized and reliable information management. Furthermore, this study seeks to demonstrate the practical application of OCR and the Levenshtein Distance Algorithm in real estate information systems and may serve as a valuable reference for future researchers and organizations pursuing intelligent document processing and integrated business information systems.

Purpose and Description

St. Francis Square Realty Corporation (SFSRC) is a private real estate company located on the 4th Floor, St. Francis Square Mall, Julia Vargas Avenue corner Bank Drive, Ortigas Center, Mandaluyong City, Metro Manila, Philippines. The company is headed by Mr. Luke C. Roxas, Chairman, and is engaged in the development, marketing, sale, and management of residential and commercial real estate properties. Over the years, the company has established itself as a trusted real estate organization by providing quality property investment opportunities and comprehensive after-sales services to its clients.

With approximately 100 employees, SFSRC serves more than 3,000 property owners and tenants, primarily within the cities of Mandaluyong and Makati. The company's operations involve several departments that work together to manage property sales, reservations, documentation, receivables, billing, loan processing, accounting, and customer service. These departments coordinate closely to ensure that

every transaction, from reservation to property turnover, is completed in accordance with company policies and regulatory requirements.

As the company's client base continues to grow, the volume of transactions and documentary requirements also increases. Consequently, SFSRC handles a substantial amount of information related to property inventories, client records, reservation documents, contracts, billing statements, payment records, and financing requirements. Efficient management of these records is essential to maintaining accurate information, ensuring timely transaction processing, and delivering quality customer service.

The current business operations of St. Francis Square Realty Corporation involve several interconnected processes that support the complete property acquisition cycle, beginning with property inquiry and ending with the transfer of ownership. While these processes have enabled the company to manage its operations effectively over the years, many activities remain dependent on manual procedures, paper-based documentation, and separate monitoring files maintained by different departments.

The reservation and sales process begins after a prospective buyer completes a property presentation and site inspection. Once the buyer decides to purchase a unit, the sales agent or accredited broker coordinates with the central office to verify the availability of the selected unit and request the corresponding payment computation. If necessary, requests for special payment terms are endorsed to management for approval. After the unit availability is confirmed and the buyer agrees to the terms and conditions, the buyer completes the reservation application and pays the required reservation fee through bank deposit, bank transfer, cash payment, or other approved payment channels.

The reservation application together with the proof of payment is submitted to the central office, where authorized personnel manually verify the documents before recording the reservation as an official sale. Buyers are required to submit all documentary requirements within thirty (30) days from the payment of the reservation fee. Once the required documents have been received and verified, the company sends confirmation through electronic mail and proceeds with the succeeding stages of the transaction. Monthly amortization payments may be settled through post-dated checks, bank deposits, online fund transfers, or over-the-counter payments at the central office. After the buyer has paid the required initial amortization, the Contract to Sell is prepared for signing by both the buyer and the authorized company representatives before notarization.

Depending on the buyer's chosen financing arrangement, the remaining balance may be settled through full cash payment, bank financing, or financing assistance from the Pag-IBIG Fund. During this stage, the company assists buyers in processing their housing loan applications and coordinating with financing institutions. Once the account has been fully paid or the housing loan proceeds have been released, the company issues a Notice of Acceptance and endorses the buyer to the Buyers Turnover Personnel for unit inspection and turnover. The transaction concludes with the execution of the Deed of Absolute Sale and the initiation of the title transfer process.

Supporting the reservation process is the company's document management procedure. After receiving the completed reservation application, the Sales Coordinator forwards the documents to the Loans Management Department for encoding. The application is subsequently routed to different departments, including Receivables,

Documentation, and Association Dues Billing. Each department manually records the client's information in its own Microsoft Excel monitoring files to support reporting and operational requirements. Physical copies of reservation documents, signed contracts, documentary requirements, and Deeds of Absolute Sale are filed manually in client folders stored inside metal filing cabinets arranged alphabetically according to condominium projects. Likewise, the Accounting Department maintains separate physical folders and electronic records for accounting purposes. Payment information and transaction progress are independently monitored by the different departments involved throughout the sales and financing process.

The receivables and collection process is likewise performed manually. Billing personnel monitor payment schedules based on the approved schedule of payments and record collections using separate monitoring files. Buyers who fail to settle two consecutive monthly obligations are issued a demand notice, while overdue accounts are charged a monthly penalty of five percent (5%). If payment remains unsettled after sixty (60) days, the account is endorsed to the company's legal counsel for cancellation proceedings. Buyers who have paid at least twenty-four (24) months of equity are entitled to a refund in accordance with applicable laws and company policies, while accounts that do not meet the required conditions may be subject to payment forfeiture. Once cancellation is finalized, the corresponding unit is returned to the company's inventory and becomes available for resale.

For completed projects, the company also manages the billing and collection of association dues. Every month, the condominium administration conducts water and electricity meter readings for each unit and prepares utility consumption reports. These

reports are forwarded to the Billing Section, where personnel manually encode the consumption data to generate the monthly Statement of Account, including association dues, utility charges, and applicable penalties. Multiple copies of the Statement of Account are prepared for distribution to the unit owner, Accounting Department, Condominium Administration, and Billing Section. Statements are delivered to clients through courier services, while payments are accepted through bank deposits, post-dated checks, or direct payments to the condominium administration. Payment confirmations are subsequently coordinated among the Cash Department, Billing Section, Accounting Department, and Condominium Administration for recording and reconciliation purposes.

Although the existing procedures enable the company to manage its operations, the reliance on manual processing creates several operational challenges. Client information, payment records, billing details, and documentary requirements are maintained separately by different departments using independent Microsoft Excel files and physical filing systems. Since the same information is repeatedly encoded by multiple personnel, the possibility of duplicate encoding, inconsistent records, and human error increases. Updating client information also requires coordination among departments, resulting in delays before all records are synchronized.

Document management presents another significant challenge. Reservation forms, contracts, proof of payment, government-issued identification cards, and other documentary requirements are manually reviewed and filed, making document retrieval time-consuming and increasing the risk of misplaced or damaged records. Employees are required to manually compare client information contained in uploaded documents

with company records before approving transactions, further increasing administrative workload and processing time.

The current payment monitoring process also requires considerable manual intervention. Clients submit proof of payment through various communication channels, after which employees manually verify the payment before updating monitoring files maintained by different departments. Likewise, Statements of Account for association dues are distributed through courier services, which may delay the receipt of billing statements and consequently affect the timeliness of client payments. These manual procedures reduce operational efficiency, prolong transaction processing, and increase the possibility of delayed customer service.

To address these operational challenges, this study proposes the development of an OCR-Based Real Estate Management System for St. Francis Square Realty Corp. Using the Levenshtein Distance Algorithm for Automated Document Validation. The proposed solution is an integrated information system composed of two interconnected platforms: a Web-Based Real Estate Portal and an Internal Management System, both operating through a centralized database.

The Web-Based Real Estate Portal serves as the company's online platform for prospective buyers and existing clients. Visitors may browse available properties and project information without registration. Prospective buyers who wish to request a tripping schedule or reserve a unit are required to create an initial account, enabling them to submit reservation requests, upload documentary requirements, upload proof of payment, and monitor the status of their reservation. Once the reservation has been officially approved, the initial account is converted into a permanent client account that

provides access to the Client Portal. Through this portal, clients can monitor their Statement of Account, billing history, payment records, uploaded documents, announcements, and other transaction-related information throughout the property acquisition process.

The Internal Management System is designed exclusively for authorized employees and implements Role-Based Access Control (RBAC) to ensure that each user can access only the modules corresponding to their assigned responsibilities. The system supports property inventory management, reservation processing, client information management, billing generation, payment monitoring, document verification, report generation, user administration, and transaction monitoring. By maintaining a centralized database, all authorized departments can access updated information in real time, eliminating duplicate encoding and improving coordination among personnel.

A key feature of the proposed system is the integration of Optical Character Recognition (OCR) and the Levenshtein Distance Algorithm for document validation. During document submission, OCR extracts relevant information from uploaded documents, while the Levenshtein Distance Algorithm compares the extracted text with records stored in the database to determine similarity. This approach enables the system to recognize minor OCR recognition errors while accurately validating client information, thereby reducing manual verification, minimizing processing errors, and improving the efficiency of document management.

The implementation of the proposed system is expected to streamline reservation processing, strengthen document validation, improve billing and payment monitoring, centralize organizational records, reduce repetitive manual encoding, and

provide clients with a more convenient and transparent online transaction experience. Ultimately, the proposed solution supports the company's continuing digital transformation by integrating customer services and internal operations into a secure, efficient, and centralized information system.

Project Context

The proposed OCR-Based Real Estate Management System for St. Francis Square Realty Corporation Using the Levenshtein Distance Algorithm for Automated Document Validation is designed to address the operational challenges identified in the company's existing business processes. The proposed solution transforms several manual procedures into an integrated digital environment while preserving the company's established policies, approval workflows, and organizational structure. Rather than replacing existing business practices, the system enhances them by automating repetitive tasks, improving information accuracy, strengthening document management, and providing efficient online services for both employees and clients.

The proposed solution consists of two interconnected platforms: the Web-Based Real Estate Portal and the Internal Management System. Both platforms are connected through a centralized database, allowing information entered in one platform to become immediately available in the other. This eliminates duplicate data entry, minimizes inconsistencies among departments, improves collaboration, and ensures that clients and employees work with the same updated information throughout the property acquisition process.

The Web-Based Real Estate Portal serves as the company's online service platform for prospective buyers and existing clients. It provides convenient online access to property information, reservation services, document submission, payment monitoring, and transaction tracking. Meanwhile, the Internal Management System serves as the company's operational platform where authorized personnel manage reservations, property inventory, client information, document verification, billing, payment monitoring, loan processing, reports, and administrative functions. Together, these two platforms create a unified information system that supports the complete real estate transaction lifecycle.

A. Web-Based Real Estate Portal Modules

1. Property Catalog Module

The Property Catalog Module serves as the public interface of the Web-Based Real Estate Portal. Visitors may browse condominium projects and available units without creating an account. The module presents project information, property descriptions, floor plans, unit specifications, pricing, amenities, parking availability, and property images, enabling prospective buyers to explore the company's offerings before initiating any transaction.

Property availability is synchronized in real time with the Internal Management System. Units that are tagged as On Hold or Sold are automatically removed from the list of available properties displayed on the portal, ensuring that buyers always view accurate and updated inventory information.

2. Initial Registration and Account Management Module

Prospective buyers who wish to request a tripping schedule or reserve a condominium unit must first create an Initial Account. This temporary account enables the company to identify prospective buyers and facilitates communication during the reservation process.

Through the Initial Account, users may:

- Schedule property tripping appointments
- Submit reservation requests
- Upload documentary requirements
- Upload proof of reservation payment
- Receive reservation notifications
- Monitor reservation status
- Update personal information

Once the reservation has been officially approved, the Initial Account is automatically converted into a Permanent Client Account, providing continuous access to the Client Portal.

3. Reservation Management Module

The Reservation Management Module enables prospective buyers to submit reservation requests online after selecting an available property.

Upon receipt of the reservation request, the system immediately changes the selected property's status to On Hold. This prevents both online buyers and authorized

personnel from reserving the same property while the reservation is undergoing evaluation. The updated inventory status is synchronized in real time between the Web-Based Real Estate Portal and the Internal Management System, eliminating the possibility of double reservations or double sales.

Once the reservation has been reviewed and officially approved by authorized personnel, the property's status automatically changes to Sold, permanently removing the unit from the list of available properties displayed on both platforms.

4. Document Management, Classification, and Automated Validation Module

The Document Management, Classification, and Automated Validation Module enables buyers to submit documentary requirements and proof of payment electronically through the Web-Based Real Estate Portal.

Unlike traditional document upload systems, the proposed solution performs two levels of automated validation before employee verification.

The first stage is Document Type Validation. Before extracting any information, the system verifies whether the uploaded document matches the document category selected by the client. Using Optical Character Recognition (OCR) together with document classification techniques, the system analyzes the document's textual content, identifying labels, and structural characteristics.

For example, if the buyer selects Birth Certificate but uploads a Driver's License, Passport, or another unrelated document, the system automatically detects the mismatch, rejects the submission, and immediately notifies the client to upload the correct document before proceeding.

After the uploaded document has been confirmed as the correct document type, the system proceeds to Information Validation. OCR extracts relevant information such as names, document numbers, identification numbers, and dates. The extracted information is then compared with the corresponding records stored in the centralized database using the Levenshtein Distance Algorithm.

Instead of relying solely on exact text matching, the algorithm evaluates the similarity between text strings, enabling the system to tolerate minor OCR recognition errors while maintaining accurate document validation.

Following automated validation, the submitted document becomes immediately available to authorized personnel for review through the Internal Management System.

Each uploaded document is assigned one of the following statuses:

- Pending
- Verified
- Approved
- Rejected
- Requires Resubmission

If the uploaded document is incorrect or fails validation, the system immediately notifies the client and requests the correct document before the transaction can proceed.

5. Client Portal Module

After reservation approval, the buyer's Initial Account is automatically converted into a Permanent Client Account.

The Client Portal serves as the buyer's personalized online workspace throughout the property acquisition process.

Through this portal, clients can:

- Monitor reservation status
- View Statements of Account (SOA)
- Review outstanding balances
- Monitor payment history
- Upload succeeding proof of payments
- Track documentary compliance
- Receive announcements and notifications
- Monitor transaction progress

Clients may also view transaction-related documents associated with their purchase, including Contracts to Sell, Deeds of Absolute Sale, Statements of Account, financing documents, Notices of Acceptance, and other company-issued records related to the transaction.

To protect the confidentiality and integrity of legal documents, these files are provided strictly for viewing purposes only. Downloading, printing, copying, screen

capturing, screen recording, or reproducing legal documents is prohibited. Clients requiring certified hard copies must personally request them from authorized company personnel in accordance with company policies.

B. Internal Management System Modules

The Internal Management System is exclusively accessible to authorized employees and supports the daily operations of the Sales, Documentation, Loans Management, Billing, Accounting, Cash, and Management Departments. The system implements Role-Based Access Control (RBAC), ensuring that employees can access only the modules and functions necessary for their assigned responsibilities.

1. Dashboard Module

Provides real-time summaries of reservations, inventory, pending document verification, billing, collections, overdue accounts, association dues, financing status, and operational statistics.

2. Property Inventory Management Module

Maintains all condominium projects, units, parking spaces, pricing, and availability.

The module automatically synchronizes property status across both platforms.

Property status follows this lifecycle:

Available → On Hold → Sold

If management approves a cancellation, the reservation is manually updated by authorized personnel, the buyer's record is retained and tagged as Cancelled, and the

property's status is restored to Available. This ensures that cancelled transactions remain searchable for auditing, reporting, and future reference while allowing the property to be offered again to prospective buyers.

3. Client Information Management Module

Maintains a centralized buyer profile containing personal information, reservation records, payment schedules, financing details, billing information, document status, association dues, and transaction history.

4. Reservation Management Module

Processes online reservations together with walk-in and broker-assisted reservations.

Authorized personnel may manually encode reservations that originate outside the Web-Based Real Estate Portal.

Following reservation approval, the system converts the buyer's Initial Account into a Permanent Client Account or automatically generates a Permanent Client Account for manually encoded reservations.

5. Document Management, Classification, and Automated Validation Module

Serves as the company's centralized Electronic Document Management System (EDMS).

The module accepts documents submitted online by clients as well as physical documents submitted directly to the office. Authorized personnel may scan and upload

hard-copy documents, allowing them to undergo the same automated document classification, OCR extraction, and Levenshtein Distance validation process.

All validated documents are stored in the buyer's centralized electronic folder, including Reservation Applications, Contracts to Sell, Deeds of Absolute Sale, Official Receipts, Notices of Acceptance, financing documents, Special Powers of Attorney, proof of payment, and other supporting documents.

This module significantly reduces reliance on physical filing systems while improving document retrieval, security, and collaboration among departments.

6. Billing Management Module

Automates the generation and monitoring of reservation fees, monthly amortizations, turnover balances, miscellaneous charges, association dues, water charges, electricity charges, penalties, discounts, and Statements of Account.

7. Payment Monitoring Module

Records payments from bank deposits, online transfers, post-dated checks, and over-the-counter transactions. Authorized personnel verify proof of payment, post payments, monitor due dates, compute penalties, and update client payment records.

8. Loan Monitoring Module

Tracks Bank Financing and Pag-IBIG Fund applications, documentary compliance, loan approval status, loan releases, turnover balances, and financing progress.

9. Reports and Analytics Module

Generates sales, reservation, inventory, billing, collection, loan monitoring, association dues, document verification, and delinquency reports to support management decision-making.

10. User and Access Management Module

Allows the System Administrator to manage employee accounts, assign user roles, control access privileges, and maintain system security through Role-Based Access Control (RBAC).

11. Audit Trail Module

Automatically records significant user activities, including reservations, document uploads, approvals, billing updates, payment postings, and administrative actions, providing a complete history for monitoring and auditing.

C. System Integration

The Web-Based Real Estate Portal and the Internal Management System operate as a single integrated solution through a centralized database. Data entered by clients or employees is synchronized in real time, ensuring consistency across all departments and user platforms.

The integration eliminates duplicate data entry, improves interdepartmental coordination, centralizes document management, strengthens inventory control, and provides a seamless transaction experience from property inquiry to turnover.

By incorporating intelligent document classification, OCR technology, the Levenshtein Distance Algorithm, centralized electronic document management, real-time inventory synchronization, secure client services, and role-based employee access, the proposed system addresses the operational challenges currently experienced by St. Francis Square Realty Corporation. It is expected to improve operational efficiency, reduce administrative workload, enhance data accuracy, strengthen document security, improve customer service, and support the company's continuing digital transformation while maintaining its existing business processes and approval policies..

Objectives of the Study

The general objective of this study is to develop an OCR-Based Real Estate Management System for St. Francis Square Realty Corporation Using the Levenshtein Distance Algorithm for Automated Document Validation that will improve the company's operational efficiency by automating and integrating its reservation processing, document management, billing, payment monitoring, and client transaction management through a centralized information system.

Specifically, the study aims to:

1. Develop an integrated Web-Based Real Estate Portal and Internal Management System connected through a centralized database to streamline real estate transactions and improve information sharing among departments.

2. Facilitate online property viewing, tripping schedule requests, reservation processing, documentary requirement submission, proof of payment upload, and transaction monitoring through the Web-Based Real Estate Portal.
3. Provide authorized personnel with an Internal Management System that supports property inventory management, client information management, reservation processing, billing, payment monitoring, loan monitoring, document management, and report generation based on assigned user roles.
4. Implement Optical Character Recognition (OCR) technology with document classification to automatically identify uploaded document types, extract relevant information, and detect incorrect documentary submissions before validation.
5. Validate the extracted document information using the Levenshtein Distance Algorithm by comparing OCR-generated data with records stored in the centralized database to improve document verification accuracy while minimizing manual encoding and validation errors.
6. Centralize the storage and management of client records and transaction-related documents by maintaining a secure electronic document repository that supports both online document submissions and scanned physical documents uploaded by authorized personnel.
7. Prevent duplicate reservations and double sales by implementing real-time property inventory synchronization between the Web-Based Real Estate Portal and the Internal Management System through automated unit status management (Available, On Hold, Sold, and Cancelled).

8. Enhance billing and payment management by automating the generation of Statements of Account, monitoring payment transactions, recording payment history, and providing clients with real-time access to their account information through the Client Portal.
9. Generate operational and management reports, including:
 - Property Inventory Report
 - Reservation Report
 - Client Information Report
 - Document Verification Report
 - Billing Report
 - Collection Report
 - Statement of Account Report
 - Payment History Report
 - Loan Monitoring Report
 - Association Dues Report
 - Delinquent Accounts Report
 - Sales Report
 - Audit Trail Report

10. Evaluate the functionality, reliability, usability, efficiency, maintainability, portability, and overall quality of the proposed system using the ISO/IEC 25010:2023 Software Quality Model.

Significance of the Study

This study is significant because it proposes an integrated OCR-Based Real Estate Management System that combines a Web-Based Real Estate Portal and an Internal Management System to improve the efficiency of real estate operations at St. Francis Square Realty Corporation. By integrating reservation processing, document management, billing, payment monitoring, and automated document validation into a centralized information system, the proposed solution aims to reduce manual processes, improve data accuracy, strengthen document security, and enhance the overall quality of customer service. The findings and implementation of this study are expected to benefit the following:

St. Francis Square Realty Corporation.

The proposed system will provide the company with an integrated digital platform that streamlines its real estate operations through automation and centralized information management. It will reduce repetitive manual tasks, minimize encoding errors, improve document retrieval, strengthen inventory control, enhance billing and payment monitoring, and facilitate better coordination among departments. The implementation of OCR technology and the Levenshtein Distance Algorithm will also improve the efficiency and accuracy of document validation, contributing to faster transaction processing and improved operational performance.

Management.

The proposed system will provide management with timely and accurate information for monitoring business operations and making informed decisions. Through automated reports, real-time operational dashboards, and centralized records, management can efficiently monitor sales, reservations, inventory, collections, document compliance, and overall business performance. The system also strengthens internal controls by maintaining audit trails and preserving complete transaction histories.

Employees and Authorized Personnel.

The Internal Management System will assist employees by automating routine tasks such as reservation processing, client information management, document verification, billing preparation, payment monitoring, and report generation. The centralized database will eliminate duplicate data entry, reduce administrative workload, improve collaboration among departments, and allow employees to perform their responsibilities more efficiently. The electronic document management feature will also simplify document storage and retrieval while reducing dependence on physical filing systems.

Clients and Prospective Buyers.

The Web-Based Real Estate Portal will provide clients and prospective buyers with convenient online access to real estate services. They will be able to browse available properties, request tripping schedules, submit reservation applications, upload

documentary requirements and proof of payment, monitor reservation status, view Statements of Account, track payment history, receive notifications, and monitor the progress of their transactions through the Client Portal. These online services will reduce the need for frequent office visits while improving communication, transparency, and overall customer experience.

Sales Agents and Brokers.

Sales agents and accredited brokers will benefit from a more efficient reservation process through faster verification of unit availability, improved coordination with the central office, and quicker processing of reservation applications. The centralized system will also reduce delays caused by manual communication and provide more accurate information regarding property inventory and transaction status.

Future Researchers.

The results of this study may serve as a useful reference for future researchers who intend to develop or enhance real estate management systems, electronic document management systems, or other business information systems utilizing Optical Character Recognition (OCR), document classification, and the Levenshtein Distance Algorithm. The study may also contribute to future research involving automated document validation, centralized information systems, and digital transformation within the real estate industry.

Academic Institutions and Information Technology Students.

This study may serve as a learning resource for students, educators, and academic institutions by demonstrating the practical application of software engineering

principles, systems analysis and design, database management, Optical Character Recognition (OCR), intelligent document validation, and algorithm implementation in solving real-world business problems. It may also provide future BSCS students with a reference for developing integrated enterprise information systems.

Overall, the proposed OCR-Based Real Estate Management System is expected to contribute to the digital transformation of St. Francis Square Realty Corporation by improving operational efficiency, strengthening document management and validation, enhancing customer service, and supporting more effective and reliable real estate transaction management.

Scope and Limitation

This study focuses on the design and development of an **OCR-Based Real Estate Management System for St. Francis Square Realty Corporation Using the Levenshtein Distance Algorithm for Automated Document Validation**. The proposed system aims to improve the company's existing business processes by integrating reservation management, document management, billing, payment monitoring, loan monitoring, client services, and operational reporting into a centralized information system. The system consists of two interconnected platforms: the **Web-Based Real Estate Portal** and the **Internal Management System**, both of which share a centralized database to ensure real-time synchronization of information and efficient collaboration among departments.

The **Web-Based Real Estate Portal** serves as the company's online platform for prospective buyers and existing clients. Visitors may browse available condominium

projects, unit information, and project details without creating an account. Prospective buyers who wish to request a tripping schedule or reserve a unit are required to register and create an **Initial Account**. Through this account, users can request property tripping, submit reservation applications, upload documentary requirements and proof of payment, receive notifications, and monitor the status of their reservation. Once the reservation has been verified and approved by authorized personnel, the Initial Account is automatically converted into a **Permanent Client Account**, allowing the buyer to access the **Client Portal**. Through the Client Portal, clients can monitor reservation status, view Statements of Account, review payment history, upload succeeding proof of payments, receive announcements and notifications, monitor documentary compliance, and securely view transaction-related documents associated with their purchase.

The **Internal Management System** is intended exclusively for authorized employees of St. Francis Square Realty Corporation. It supports the operational functions of the Sales, Documentation, Loans Management, Billing, Accounting, Cash, and Management Departments through modules for property inventory management, client information management, reservation processing, document management, billing, payment monitoring, loan monitoring, report generation, user management, and audit trail monitoring. The system implements **Role-Based Access Control (RBAC)** to ensure that users can access only the modules and functions corresponding to their assigned responsibilities.

The proposed system automates document management through a **two-stage document validation process**. Documentary requirements may be uploaded in **PDF, JPG, JPEG, or PNG** format with a maximum file size of **3 MB** per document.

Documents may be submitted either as scanned copies or as clear photographs captured using a mobile device or digital camera.

During the first stage, the system performs **Document Type Validation**. Users are required to select the document category before uploading a file, such as a Birth Certificate, Marriage Certificate, Valid Government-Issued Identification Card, Proof of Billing, Special Power of Attorney, or other required documents. The system then analyzes the uploaded document by extracting its text through Optical Character Recognition (OCR) and comparing recognizable document titles, keywords, labels, and standard document structures against predefined document patterns stored within the system. If the uploaded document does not correspond to the selected document category, the system immediately notifies the user that an incorrect document has been uploaded and requests the submission of the appropriate document before processing can continue.

Once the document type has been successfully verified, the second stage performs **information validation**. The OCR component extracts relevant information, such as the client's name, document number, dates, and other required details. The extracted information is then compared with the corresponding records stored in the centralized database using the **Levenshtein Distance Algorithm**, which measures the similarity between text strings and tolerates minor differences caused by OCR recognition errors. This process improves the accuracy of automated document validation while significantly reducing manual data encoding and verification.

The system supports both online document submission through the Web-Based Real Estate Portal and manual document uploading by authorized personnel for buyers

who submit physical documents directly to the company or through accredited sales agents and brokers. All validated documents, including reservation forms, documentary requirements, Contracts to Sell, Deeds of Absolute Sale, proof of payment, financing documents, and other transaction-related records, are stored in a centralized electronic document repository, providing a complete digital record for every buyer throughout the property acquisition process.

The study also covers the automation of reservation processing and property inventory monitoring. Once a reservation request is accepted for processing, the selected unit is automatically tagged as **"On Hold"**, preventing multiple buyers from reserving the same property. This reservation status is synchronized in real time between the Web-Based Real Estate Portal and the Internal Management System to eliminate double-selling and maintain accurate property availability. Once the reservation has been officially approved, the unit status automatically changes to **"Sold"**, making it unavailable for future reservations.

If management subsequently approves the cancellation of a reservation, authorized personnel may manually process the cancellation through the Internal Management System. The buyer's transaction record is retained within the database and tagged as **"Cancelled"** rather than being permanently deleted, preserving the complete transaction history for auditing, reporting, and future reference. Upon completion of the approved cancellation process, the associated property is automatically restored to **"Available"** status and becomes visible once again within the property inventory.

The proposed system also automates billing and payment monitoring by generating Statements of Account, recording payment transactions, monitoring payment history, and generating operational reports, including reservation reports, inventory reports, billing reports, collection reports, document verification reports, loan monitoring reports, association dues reports, sales reports, delinquent account reports, and audit trail reports. Clients may access their financial information through the Client Portal, while authorized personnel manage billing and payment activities through the Internal Management System.

To protect confidential information and legal documents, the Client Portal provides **view-only access** to transaction-related records. Clients may view Contracts to Sell, Deeds of Absolute Sale, Statements of Account, financing documents, Notices of Acceptance, and other company-issued documents associated with their transactions. However, downloading, printing, copying, screen capturing, screen recording, or otherwise reproducing these documents through the system is prohibited. Clients requiring certified hard copies must personally request them from authorized company personnel in accordance with the company's existing policies.

Limitations of the Study

The proposed system is developed specifically for the operational requirements of **St. Francis Square Realty Corporation** and follows the company's existing business policies, procedures, and organizational structure. It is not intended for implementation in other organizations without further modification and customization.

The **Internal Management System** is designed as an **office-based information system** that operates only within the company's **Local Area Network (LAN)**. All transaction records, client information, and system data are stored in a centralized database hosted on the company's server located in the **Information Technology (IT) Department**. Consequently, the Internal Management System cannot be installed on personal computers or laptops for use outside the company's premises, nor can employees remotely access its operational modules from external locations. This implementation ensures centralized data management, maintains database integrity, and protects confidential company information.

In contrast, the **Web-Based Real Estate Portal** is accessible through the Internet but is limited to client-oriented services. Its functions include property viewing, Initial Account registration, tripping requests, reservation applications, documentary requirement submission, proof of payment upload, reservation monitoring, Client Portal access, viewing Statements of Account, reviewing payment history, receiving notifications, and viewing transaction-related documents. Administrative functions such as reservation approval, client record management, inventory management, billing generation, payment posting, document verification, report generation, user administration, and system configuration are available only through the Internal Management System and may be performed solely by authorized personnel.

The proposed system does not include a mobile application. All client services are provided through the Web-Based Real Estate Portal using standard web browsers, while employee operations are performed exclusively through desktop or laptop computers connected to the company's internal network.

The document validation component is intended to assist authorized personnel and is not designed to replace human judgment. Although the system employs **Document Type Validation**, OCR technology, and the **Levenshtein Distance Algorithm** to improve document verification accuracy, recognition performance depends on the quality of the uploaded documents. Documents that are blurred, cropped, damaged, captured under poor lighting conditions, or scanned at very low resolutions may reduce OCR accuracy. Likewise, document type validation depends on the presence of recognizable document titles, keywords, and standard document layouts. Documents with incomplete content, missing pages, or heavily modified formats may not be correctly classified. Therefore, the final approval or rejection of documentary requirements remains the responsibility of authorized personnel.

Furthermore, the proposed system does not support online contract signing, electronic notarization, online payment gateway integration, automated loan approval by accredited banks or the Pag-IBIG Fund, title transfer processing, or direct integration with external government agencies and financial institutions. These activities remain outside the scope of the study and continue to follow the company's existing operational procedures and approval processes.

During the development of the proposed system, the researchers recognize that the effectiveness of OCR-based validation depends on document quality, while the performance of the Levenshtein Distance Algorithm depends on the accuracy of the OCR-extracted text. As a result, the proposed system is intended to serve as a decision-support tool that enhances operational efficiency and minimizes manual encoding rather than completely automating all business decisions. Critical activities

such as reservation approval, document acceptance, payment verification, account cancellation, and other major business transactions remain subject to company policies and the approval of authorized personnel.

Definition of Terms

The following terms are defined conceptually and operationally as used in this study to facilitate a better understanding of the proposed system.

Audit Trail. A system-generated record that logs user activities performed within the Internal Management System, including logins, record creation, updates, approvals, and other transactions, for monitoring, accountability, and security purposes.

Centralized Database. A single database repository used by both the Web-Based Real Estate Portal and the Internal Management System to store and manage client records, property information, reservations, billing, payments, documentary requirements, and transaction history in real time.

Client Portal. The secured section of the Web-Based Real Estate Portal that becomes accessible after a buyer's reservation has been approved. It enables clients to monitor reservation status, view Statements of Account, review payment history, upload succeeding proof of payments, receive announcements, and view transaction-related documents associated with their purchase.

Document Type Validation. The initial document verification process performed by the proposed system to determine whether the uploaded document corresponds to the document category selected by the user. The system analyzes document titles, keywords, labels, and structural patterns before allowing further document processing.

Electronic Document Repository. A centralized digital storage facility within the proposed system where validated documentary requirements, contracts, proof of payment, financing documents, and other transaction-related records are securely stored and managed throughout the buyer's transaction lifecycle.

Initial Account. A temporary user account created by a prospective buyer to request a tripping schedule, submit a reservation application, upload documentary requirements and proof of payment, and monitor the reservation process. Upon reservation approval, the Initial Account is automatically converted into a Permanent Client Account.

Internal Management System. The office-based information system developed exclusively for authorized employees of St. Francis Square Realty Corporation. It manages reservation processing, property inventory, client records, document management, billing, payment monitoring, loan monitoring, report generation, user administration, and other operational functions.

Levenshtein Distance Algorithm. A string similarity algorithm that measures the minimum number of character insertions, deletions, or substitutions required to transform one text string into another. In this study, it is used to compare OCR-extracted information with records stored in the centralized database to improve document validation accuracy despite minor recognition errors.

Optical Character Recognition (OCR). A document recognition technology that extracts machine-readable text from scanned documents and digital images. In this study, OCR is used to retrieve relevant information from uploaded documentary requirements before validation.

Permanent Client Account. The official client account automatically generated after the buyer's reservation has been approved. It provides access to the Client Portal and allows clients to monitor their transactions and view information related to their purchased property.

Property Inventory Management. The module responsible for maintaining and updating records of available, reserved, on-hold, sold, and cancelled condominium units to ensure accurate property availability throughout the reservation process.

Real-Time Synchronization. The automatic updating of information between the Web-Based Real Estate Portal and the Internal Management System whenever a transaction is processed, ensuring that both platforms display consistent and current data.

Reservation Management. The process of receiving, verifying, approving, monitoring, and recording reservation applications while updating the status of property units and buyer accounts within the proposed system.

Role-Based Access Control (RBAC). A security mechanism that restricts system access according to the assigned roles and responsibilities of authorized employees, ensuring that users can access only the modules and functions necessary for their work.

Statement of Account (SOA). A system-generated document summarizing a client's financial obligations, including amortization schedules, association dues, utility charges, penalties, payments made, outstanding balances, and other related charges.

Transaction History. The complete chronological record of a buyer's reservation, payments, documentary submissions, approvals, billing activities, loan processing, and other transactions maintained within the proposed system.

Web-Based Real Estate Portal. The Internet-accessible platform developed for prospective buyers and existing clients of St. Francis Square Realty Corporation. It enables property viewing, tripping requests, reservation applications, document submission, payment monitoring, and access to client-related services through the Client Portal.

Technical Terms

The following technical terms are conceptually defined based on authoritative references and are arranged alphabetically.

Algorithm. A finite sequence of well-defined computational steps used to solve a problem or perform a specific task (Cormen et al., 2022).

Database. An organized collection of related data designed to support efficient storage, retrieval, modification, and management of information (Elmasri & Navathe, 2017).

Document Management System (DMS). A computer-based system used to capture, store, organize, manage, retrieve, and track electronic documents and digital records throughout their life cycle (International Organization for Standardization [ISO], 2016).

Joint Photographic Experts Group (JPEG/JPG). A commonly used digital image format that employs lossy compression to reduce file size while maintaining acceptable image quality for storage and transmission (International Organization for Standardization/International Electrotechnical Commission [ISO/IEC], 1994).

Local Area Network (LAN). A communication network that interconnects computers and other devices within a limited geographic area, such as an office building or organization, to facilitate data sharing and resource access (Stallings, 2020).

Portable Document Format (PDF). A file format developed to present and exchange documents consistently across different software applications, hardware platforms, and operating systems (Adobe, n.d.).

Portable Network Graphics (PNG). A raster graphics file format that supports lossless data compression and is widely used for storing high-quality digital images (World Wide Web Consortium [W3C], 2003).

System Integration. The process of connecting different software applications, subsystems, or technologies so they function together as a unified system and share information seamlessly (IBM, n.d.).

Web-Based Information System. An information system that delivers services, manages data, and provides system functionality through web technologies accessible using standard web browsers over a network (Turban et al., 2018).

Workflow Automation. The use of technology to automate business processes by executing predefined tasks, routing information, and minimizing manual intervention to improve efficiency and consistency (Microsoft, n.d.).

Operational Terms

The following terms are operationally defined based on how they are used in this study.

Account Activation refers to the automatic conversion of a buyer's Initial Account into a Permanent Client Account after the reservation has been reviewed and approved by authorized personnel.

Available Unit refers to a condominium unit that has not been reserved, sold, or placed on hold and is displayed as open for reservation through the Web-Based Real Estate Portal.

Cancelled Transaction refers to a reservation or purchase transaction that has been officially cancelled upon management approval. The transaction record remains stored in the system for auditing, reporting, and future reference, while the associated unit is returned to the available inventory.

Document Classification refers to the process of determining whether the uploaded document matches the document category selected by the user before information extraction and validation are performed.

Document Compliance Status refers to the system-generated indicator showing whether a buyer has submitted all required documentary requirements, has pending documents, or has incomplete submissions.

Document Verification Result refers to the outcome of the automated document validation process, indicating whether the uploaded document passed document classification and information validation or requires further review by authorized personnel.

Electronic Client Folder refers to the centralized digital repository containing all records related to a buyer, including documentary requirements, contracts, proof of payments, Statements of Account, and other transaction-related documents.

On Hold Status refers to the temporary status assigned to a property unit immediately after a reservation request has been accepted for processing. A unit with this status cannot be reserved by another prospective buyer until the reservation is approved or cancelled.

Payment Verification refers to the process of confirming that the proof of payment submitted by the buyer corresponds to the payment information recorded by the company before the payment is officially posted to the buyer's account.

Real-Time Inventory Update refers to the automatic synchronization of property availability between the Web-Based Real Estate Portal and the Internal Management System whenever a reservation is submitted, approved, or cancelled.

Reservation Queue refers to the list of reservation applications awaiting verification and approval by authorized personnel.

Sold Status refers to the property status assigned to a unit after the reservation has been officially approved, making the unit unavailable for future reservations in both interconnected platforms.

Transaction Monitoring refers to the continuous tracking of a buyer's reservation progress, document submission, billing, payments, loan processing, and other related activities from reservation until completion of the transaction.

View-Only Document Access refers to the security feature that allows clients to view company-issued documents through the Client Portal without permitting downloading, printing, copying, screen capturing, or screen recording. Requests for certified hard copies must be made personally through authorized company personnel.

Workstation-Based Access refers to the restriction that allows the Internal Management System to operate only on authorized computers connected to the company's Local Area Network (LAN), with all information stored in the centralized server maintained by the Information Technology Department.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

Introduction

The real estate industry has undergone significant transformation due to the rapid advancement of information and communication technologies. Traditional business operations that relied heavily on manual documentation, paper-based records, and face-to-face transactions are gradually being replaced by integrated digital systems that improve operational efficiency, enhance customer experience, and support data-driven decision-making. These technological developments have encouraged real estate organizations to adopt web-based platforms, electronic document management systems, intelligent document processing, and automated transaction monitoring to remain competitive and responsive to changing customer expectations.

For St. Francis Square Realty Corporation, several business processes—including reservation processing, documentary verification, billing, payment monitoring, and client record management—are still partially dependent on manual procedures. These processes require employees to encode client information repeatedly, maintain multiple copies of documents across different departments, and manually verify documentary requirements submitted by buyers. Such practices increase processing time, create duplicate records, and expose the organization to risks associated with misplaced documents, inconsistent data, and delayed transaction processing.

To address these challenges, the proposed study aims to develop an **OCR-Based Real Estate Management System for St. Francis Square Realty Corporation**

Using the Levenshtein Distance Algorithm for Automated Document Validation.

The system integrates a **Web-Based Real Estate Portal** for prospective buyers and existing clients with an **Internal Management System** used by authorized personnel. Both platforms utilize a centralized database that supports real-time synchronization of reservation records, document management, billing information, payment monitoring, and client transactions.

This chapter presents the literature and studies that support the development of the proposed system. It discusses current concepts, technologies, and previous research related to digital transformation, web-based information systems, electronic document management, Optical Character Recognition (OCR), document validation, role-based access control, and real estate information systems. The reviewed literature provides the theoretical and technological foundation of the study, while the reviewed studies identify research gaps that justify the need for the proposed system.

Related Literature

Digital Transformation in the Real Estate Industry (Foreign Literature)

Digital transformation has become one of the primary drivers of innovation in the real estate industry. Traditionally, property developers and real estate organizations relied on manual workflows involving paper-based documentation, physical property marketing, and face-to-face transactions. However, advances in digital technologies have enabled organizations to redesign their operational processes by integrating intelligent information systems, cloud computing, artificial intelligence, data analytics, Internet of Things (IoT), and web-based customer services. These technologies not only

improve operational efficiency but also enhance customer engagement, increase organizational transparency, and support informed decision-making.

Al-haimi et al. (2025) conducted a systematic literature review of digital transformation in the real estate industry using the PRISMA methodology. The review analyzed 36 scholarly publications and found that digital technologies have fundamentally changed property management, customer interaction, investment analysis, and operational processes. The study identified several benefits of digital transformation, including faster business processes, improved decision-making, enhanced customer experience, and better utilization of organizational resources. Nevertheless, the authors also noted challenges such as implementation costs, cybersecurity concerns, organizational resistance to change, and integration of legacy systems.

Similarly, Harth, Maqbool, and Gledson (2025/2026) introduced the concept of **Real Estate 4.0**, describing how emerging technologies such as artificial intelligence, cloud computing, big data, and digital business models are reshaping the real estate sector. Their review highlighted the importance of system integration and digital infrastructure in supporting modern real estate operations. The authors emphasized that organizations should move beyond isolated software solutions and instead adopt integrated platforms capable of supporting multiple business functions within a centralized environment.

The literature also indicates that customers increasingly expect digital services throughout the property acquisition process. Rather than visiting company offices multiple times, buyers prefer online access to property information, reservation status,

payment history, and transaction updates. Consequently, many real estate organizations are investing in web portals and customer self-service platforms that improve accessibility while reducing administrative workload.

The reviewed literature supports the proposed study because St. Francis Square Realty Corporation currently performs several business processes manually, including reservation management, document handling, payment monitoring, and client record management. The proposed OCR-Based Real Estate Management System aligns with current digital transformation initiatives by integrating these business functions into a centralized platform composed of a **Web-Based Real Estate Portal** and an **Internal Management System**. Through digital transformation, the company can reduce repetitive manual processes, improve operational efficiency, strengthen information management, and enhance customer service.

Web-Based Information Systems (Foreign Literature)

Web-based information systems have become an essential component of modern organizations because they enable users to access organizational services through standard web browsers without requiring specialized software installation. Unlike traditional desktop applications, web-based systems provide centralized data management, real-time information sharing, and remote accessibility for authorized users. These characteristics make web-based systems particularly suitable for organizations managing large volumes of transactions involving multiple departments and external clients.

According to recent literature on digital real estate, web-based technologies have significantly transformed property transactions by enabling organizations to provide online property catalogs, reservation systems, customer portals, electronic payment monitoring, and digital document management. Digital platforms improve communication between organizations and their clients while minimizing delays associated with manual transaction processing. Furthermore, centralized web applications facilitate data consistency because all users access a common database containing updated information.

Digital real estate platforms also contribute to improved customer satisfaction by providing self-service capabilities. Through web portals, prospective buyers can browse available properties, submit reservation requests, upload documentary requirements, receive transaction updates, and monitor payment information without repeatedly visiting the company's office. These capabilities improve convenience while reducing the administrative workload of company personnel.

However, researchers emphasize that web-based information systems handling confidential business information should implement appropriate security mechanisms, including user authentication, access control, encrypted communication, and centralized database management. Organizations processing legal documents and financial records must ensure that sensitive information remains accessible only to authorized users. Proper implementation of role-based permissions and secure document viewing is therefore essential in protecting organizational data.

The reviewed literature directly supports the proposed study. The **Web-Based Real Estate Portal** allows prospective buyers to browse available condominium units

without creating an account. Once a buyer decides to reserve a property, an Initial Account is created to facilitate reservation requests, document submission, and proof-of-payment uploads. Upon reservation approval, the account is converted into a Permanent Client Account that provides secure access to reservation status, Statements of Account, payment history, announcements, documentary compliance, and transaction-related documents. Meanwhile, the **Internal Management System** supports the operational activities of authorized personnel within the company through a centralized platform connected to the organization's internal server. This dual-platform architecture reflects current practices in web-based information system development while addressing the specific operational requirements of St. Francis Square Realty Corporation.

Related Literature

Electronic Document Management Systems (Foreign Literature)

Electronic Document Management Systems (EDMS) have become an essential component of modern organizations because they enable the systematic creation, storage, retrieval, management, and preservation of digital documents throughout their lifecycle. Unlike traditional paper-based filing systems, EDMS centralizes organizational records in a secure electronic repository, allowing authorized users to retrieve documents quickly while minimizing the risks of document loss, duplication, and unauthorized access. Organizations adopting electronic document management benefit from improved operational efficiency, enhanced collaboration among departments, reduced storage costs, and stronger compliance with organizational policies and regulatory requirements. Recent studies further emphasize that digital document

repositories improve organizational productivity by reducing manual document handling and accelerating document retrieval processes.

Modern electronic document management systems no longer function merely as digital filing cabinets. They now incorporate workflow automation, metadata management, version control, audit trails, role-based permissions, and intelligent document processing technologies to support end-to-end business operations. These capabilities enable organizations to monitor document movement, maintain document integrity, and ensure that only authorized personnel can create, modify, approve, or access confidential records. As organizations continue to digitize their operations, centralized document management has become an important strategy for improving information governance and organizational efficiency.

Another important characteristic of modern document management systems is the integration of electronic document repositories with operational information systems. Rather than maintaining separate databases for each department, organizations increasingly utilize centralized repositories that allow multiple departments to access the same validated information while maintaining appropriate access restrictions. Such integration minimizes duplicate encoding, improves data consistency, and enhances interdepartmental coordination.

The reviewed literature strongly supports the proposed study. At present, St. Francis Square Realty Corporation stores reservation documents, documentary requirements, Contracts to Sell, Deeds of Absolute Sale, Statements of Account, proof of payments, and financing documents in separate physical folders maintained by different departments. This practice increases document redundancy, retrieval time, and

the possibility of inconsistent records. The proposed **OCR-Based Real Estate Management System** addresses these issues by establishing a centralized electronic document repository where all validated client documents are stored in a single electronic client folder. Regardless of whether documents are submitted online through the Web-Based Real Estate Portal or uploaded manually by authorized personnel after receiving physical copies from clients, all documents become part of the client's centralized electronic record. This approach supports faster document retrieval, improved record management, and better coordination among the Sales, Loans Management, Billing, Accounting, Cash, and Management Departments.

Optical Character Recognition (OCR) (Foreign Literature)

Optical Character Recognition (OCR) is one of the most widely used technologies for converting printed or scanned documents into machine-readable text. The technology enables computer systems to identify characters from scanned images, photographs, and Portable Document Format (PDF) files, thereby eliminating the need for repetitive manual data encoding. OCR has become an important component of digital transformation because it significantly improves document processing efficiency while reducing human encoding errors. It is widely applied in government agencies, banking institutions, healthcare organizations, educational institutions, logistics companies, and business enterprises that process large volumes of documents.

Recent advancements in OCR technology have expanded its capabilities beyond simple text extraction. Modern OCR engines are capable of recognizing complex page layouts, tables, forms, and structured documents while preserving their spatial relationships. Deep learning techniques and transformer-based OCR models have

significantly improved recognition accuracy, particularly when processing machine-printed documents with varying layouts and formats. Nevertheless, researchers acknowledge that OCR performance remains dependent on document quality, image resolution, lighting conditions, scanning accuracy, and text clarity. Documents containing blurred images, damaged pages, poor lighting, handwritten text, or distorted characters may still produce recognition errors.

Current literature also emphasizes that OCR should not be viewed as a complete replacement for human document verification. Instead, OCR serves as an intelligent support tool that accelerates information extraction while allowing authorized personnel to review extracted information before final approval. Organizations handling legal and financial documents continue to combine OCR technology with validation algorithms and human verification to ensure document authenticity and accuracy.

The reviewed literature is highly relevant to the proposed study because the system utilizes OCR as the primary technology for automating documentary verification. Buyers may submit documentary requirements in **PDF, JPG, JPEG, or PNG** formats with a maximum file size specified by the company. After upload, OCR extracts relevant textual information such as the owner's name, document number, birth date, and other required fields. However, the proposed system does not rely solely on OCR results. Instead, extracted information is further validated using the Levenshtein Distance Algorithm before being reviewed by authorized personnel. Consequently, OCR serves as an intelligent document processing tool that reduces repetitive manual encoding while maintaining human supervision during document approval.

Intelligent Document Processing (Foreign Literature)

Intelligent Document Processing (IDP) represents the evolution of traditional document management by combining Optical Character Recognition with Artificial Intelligence, Natural Language Processing, Machine Learning, and automated validation techniques. Unlike conventional OCR systems that merely convert images into editable text, IDP interprets document content, classifies document types, extracts relevant information, validates extracted data, and integrates the processed information into organizational workflows. As a result, IDP enables organizations to automate document-intensive business processes while maintaining high levels of accuracy and operational efficiency.

Recent literature indicates that intelligent document processing has become increasingly important in industries requiring extensive document verification, including banking, insurance, healthcare, logistics, government services, and real estate. By integrating OCR with intelligent validation techniques, organizations reduce repetitive administrative work, improve processing speed, and enhance data quality. Modern IDP solutions also perform document classification before information extraction, ensuring that uploaded files correspond to the expected document category before further processing. This preliminary validation reduces processing errors and prevents incorrect documents from entering organizational databases.

The proposed study adopts this concept through the implementation of **Document Type Validation**. Before OCR begins extracting textual information, the system first verifies whether the uploaded document matches the document type selected by the client. For example, if the client selects **Birth Certificate** but uploads a driver's license instead, the system immediately identifies the inconsistency and notifies

the user to upload the correct document. Only after successful document type verification does the system proceed with OCR-based text extraction and subsequent validation. This approach improves data quality while minimizing unnecessary processing of incorrect documentary submissions.

Levenshtein Distance Algorithm (Foreign Literature)

The Levenshtein Distance Algorithm is a widely recognized string similarity algorithm used to measure the minimum number of insertions, deletions, or substitutions required to transform one text string into another. Rather than requiring exact character matching, the algorithm determines the degree of similarity between two strings, making it particularly useful in applications involving spelling correction, duplicate record detection, fuzzy searching, data cleaning, and document validation. Recent research on document understanding continues to use Levenshtein-based similarity measures when evaluating OCR performance because they effectively account for minor recognition errors without treating every mismatch as a completely different value.

Within OCR-based systems, character recognition errors may occur because of image quality, scanning resolution, lighting conditions, or font variations. For example, OCR may recognize the surname **"Cruz"** as **"Crvz"** or **"Dela"** as **"De1a."** Traditional exact matching algorithms would identify these values as different, resulting in failed validation. In contrast, the Levenshtein Distance Algorithm calculates the similarity between the two strings and recognizes that only a minimal number of character substitutions exists. This capability improves validation accuracy while reducing false mismatches generated by OCR processing.

In the proposed study, the Levenshtein Distance Algorithm is not intended to determine whether a document is authentic. Instead, its primary function is to compare OCR-extracted information with the expected information stored in the company's centralized database. If the similarity score falls within the predefined acceptable threshold established during system development, the document is considered a potential match and forwarded to authorized personnel for final verification. Otherwise, the system flags the discrepancy for further review. Consequently, the algorithm serves as a decision-support mechanism that assists employees during documentary validation rather than replacing human judgment.

Role-Based Access Control (RBAC) (Foreign Literature)

As organizations increasingly adopt integrated information systems, protecting sensitive organizational data has become a major concern. Modern information systems process confidential business records, financial transactions, legal documents, and customer information that should only be accessible to authorized personnel. One of the most widely adopted security mechanisms for addressing this concern is **Role-Based Access Control (RBAC)**, which restricts system access according to the responsibilities assigned to each user within an organization.

RBAC operates on the principle of **least privilege**, where users are granted only the permissions necessary to perform their assigned duties. Instead of assigning permissions individually to every employee, the system assigns permissions to predefined organizational roles such as administrator, manager, cashier, billing personnel, or documentation staff. Users inherit the permissions associated with their assigned role, making access management more efficient, secure, and easier to

maintain. RBAC has become the standard authorization model used in enterprise information systems because it simplifies permission administration while improving accountability and protecting confidential organizational information.

Recent studies further demonstrate that modern RBAC implementations now support hierarchical roles, fine-grained permissions, and context-aware authorization to accommodate complex organizational structures. Wang (2026) proposed the Zoned Role-Based (ZRB) Framework, which extends the traditional RBAC model by integrating organizational hierarchy directly into the system architecture. The study emphasized that enterprise information systems should reflect the actual organizational structure so that permissions, responsibilities, and business processes remain consistent across different departments.

Similarly, Maji, Mandal, and Mitra (2026) introduced **Encrypted Role-Based Access Control (ERBAC)**, which combines RBAC with encryption and row-level database security to strengthen data protection within enterprise systems. Their findings highlight that combining RBAC with secure database management significantly reduces unauthorized access while maintaining efficient system performance.

The reviewed literature strongly supports the proposed study. Since St. Francis Square Realty Corporation involves multiple departments—including Sales, Loans Management, Documentation, Billing, Accounting, Cash, Condominium Administration, Management, and the IT Department—the proposed Internal Management System will implement **Role-Based Access Control** to ensure that each employee can only access the modules necessary for their assigned responsibilities.

For example:

- Sales personnel may manage reservations and buyer information.
- Documentation personnel may validate documentary requirements and upload signed legal documents.
- Billing personnel may generate Statements of Account and monitor collections.
- Cash personnel may record payments.
- Management may access executive reports and dashboards.
- The IT Department will administer the centralized database and user accounts.

By implementing RBAC, the proposed system strengthens information security, improves accountability through audit trails, and minimizes unauthorized access to confidential client records and financial information.

Centralized Database Management Systems (Foreign Literature)

A centralized database is a database architecture in which organizational information is stored and managed within a single repository that is shared by multiple users and departments. Rather than maintaining separate databases for different offices or business units, centralized database systems ensure that all authorized users access the same updated information, thereby reducing duplicate records, inconsistent data, and synchronization problems.

According to a recent systematic review on authorization and database models, centralized databases remain the preferred architecture for enterprise information systems because they improve data consistency, simplify backup and recovery

procedures, facilitate security implementation, and provide a single source of truth for organizational information. The review further emphasized that centralized databases work effectively when combined with strong access control mechanisms such as RBAC and detailed audit logging.

Modern enterprise systems increasingly integrate centralized databases with multiple applications, allowing departments to collaborate using synchronized information in real time. This architecture enables organizations to eliminate repetitive encoding because information entered by one department immediately becomes available to other authorized departments. Consequently, centralized databases improve operational efficiency, reduce processing delays, and strengthen organizational decision-making through consistent and reliable information.

The proposed study adopts this architecture by utilizing a **centralized database located on the company's internal server maintained by the IT Department**. Both the **Web-Based Real Estate Portal** and the **Internal Management System** access the same centralized database, allowing information to be synchronized in real time.

For example:

- Once a buyer submits a reservation online, the reservation immediately appears in the Internal Management System.
- Once authorized personnel approve the reservation, the buyer's Initial Account is automatically converted into a Permanent Client Account.
- When documentary requirements are validated, all departments immediately access the updated client record.

- Billing information, payment records, loan status, and reservation history are synchronized without requiring duplicate encoding.

The centralized database also supports the company's policy that the Internal Management System can only be accessed within the company's office network. Since the database resides on the organization's internal server, authorized employees cannot install the Internal Management System on personal computers for use outside the office. This configuration enhances security while ensuring centralized control of organizational information.

Customer Self-Service Portals (Foreign Literature)

Customer self-service portals have become an important component of modern business information systems because they allow customers to perform transactions independently through online platforms. Rather than relying solely on face-to-face transactions or telephone inquiries, organizations now provide web portals where customers can access services, monitor transactions, submit documents, and communicate with company personnel at their convenience.

Recent digital transformation studies indicate that customer portals improve service quality by reducing response time, increasing transparency, and providing customers with continuous access to transaction information. Self-service portals also reduce the administrative workload of employees because customers can independently perform routine activities such as monitoring account status, submitting documentary requirements, viewing billing information, and receiving system notifications.

However, organizations handling legal documents and confidential financial records must implement appropriate security mechanisms within customer portals. Sensitive documents should remain protected from unauthorized distribution through authentication, access control, encrypted communication, and document viewing restrictions. These measures help preserve document integrity while complying with organizational policies regarding confidential records.

The proposed **Web-Based Real Estate Portal** adopts these principles by providing prospective buyers and existing clients with secure online access to company services. Prospective buyers may browse available condominium projects without creating an account. To submit reservation requests, upload documentary requirements, or schedule property trippings, users must first create an Initial Account.

Once the reservation has been approved by authorized personnel, the account is automatically converted into a Permanent Client Account, allowing clients to:

- monitor reservation status;
- view Statements of Account;
- review payment history;
- upload succeeding proof of payments;
- monitor documentary compliance;
- receive announcements and transaction notifications; and
- **view** contracts, financing documents, Statements of Account, and other transaction-related records associated with their purchase.

To protect confidential legal documents, the Client Portal provides **view-only access**. Clients cannot download, print, copy, screen capture, or screen record company-issued documents through the system. Certified hard copies remain available only upon personal request from authorized company personnel, consistent with the existing policies of St. Francis Square Realty Corporation.

Local Literature

Digital Transformation in the Philippines

The Philippine government continues to promote digital transformation across both public and private sectors through initiatives that encourage organizations to modernize business processes, improve service delivery, and strengthen data management. Digital transformation aims to replace fragmented manual procedures with integrated information systems that increase operational efficiency, transparency, and accessibility.

Recent Philippine implementations demonstrate that centralized web-based information systems significantly reduce document processing time, improve information accuracy, and enhance user satisfaction by providing integrated services supported by centralized databases and secure role-based access.

These initiatives support the direction of the proposed study. By replacing manual reservation processing, document management, billing, payment monitoring, and client record management with an integrated OCR-Based Real Estate Management System, St. Francis Square Realty Corporation aligns its operations with current digital transformation practices in the Philippines.

Electronic Records Management in Philippine Organizations

Electronic records management has become increasingly important among Philippine organizations seeking to improve operational efficiency while protecting organizational records. Recent local studies emphasize that centralized document management systems reduce processing delays, minimize document duplication, strengthen information security, and improve collaboration among organizational units.

Bumalod and Velasco (2024) demonstrated that integrating document management with **Role-Based Access Control**, document classification, and encryption significantly improves both operational efficiency and information security within organizational document repositories. Their study recommends centralized document storage combined with controlled user permissions as an effective strategy for protecting sensitive organizational information.

These findings directly support the proposed study because all buyer-related documents—including reservation forms, documentary requirements, Contracts to Sell, Deeds of Absolute Sale, financing documents, and Statements of Account—will be stored in a centralized electronic client folder while remaining accessible only to authorized personnel according to their assigned roles.

Related Studies

Foreign Related Studies

OCR-Based Intelligent Document Processing Systems

Recent advancements in Intelligent Document Processing (IDP) have enabled organizations to automate document-intensive business processes through the integration of Optical Character Recognition (OCR), artificial intelligence, and document classification technologies. Lin, Dias, and Correia (2025) developed a document data matching framework for blockchain-supported real estate transactions. Their proposed system combined OCR, Natural Language Processing (NLP), and automated document matching to extract and verify information from heterogeneous real estate documents. The study demonstrated that integrating OCR with automated validation significantly reduced document verification time while improving consistency and transparency throughout real estate transactions.

The findings support the proposed study because both systems utilize OCR to automate document processing. However, the proposed OCR-Based Real Estate Management System differs by focusing on the operational workflow of St. Francis Square Realty Corporation. Rather than integrating blockchain technology, the proposed system combines **Document Type Validation**, **OCR**, and the **Levenshtein Distance Algorithm** to verify client-submitted documentary requirements before final approval by authorized personnel. The study also extends automation beyond document verification by integrating reservation processing, billing, payment monitoring, and centralized document management.

Document Management Systems with OCR Integration

Chinnasamy et al. (2025) developed a blockchain-based electronic document management system with Role-Based Access Control (RBAC) for managing educational records. The system incorporated OCR technology for automated document processing while employing RBAC and encryption techniques to protect confidential information. Evaluation results demonstrated improvements in document security, integrity, and operational efficiency.

Similarly, Sison et al. (2024) proposed a blockchain-based document management system integrated with OCR to automate document handling and verification. The system reduced manual workload, improved document retrieval, and achieved high OCR accuracy while complying with ISO/IEC 25010 software quality standards.

These studies demonstrate the effectiveness of integrating OCR with document management systems. However, both focus primarily on document digitization and secure storage. In contrast, the proposed study applies OCR within the broader context of real estate management. The system not only digitizes documents but also validates document type, compares extracted information using the Levenshtein Distance Algorithm, and stores all validated records in a centralized electronic client folder shared across multiple organizational departments.

Property Management Systems

Encarnacion, Raz, and Araneta (2025) developed an advanced Property Management System using Unified Object-Oriented Analysis and Design (OOAD) to

improve operational efficiency within property management organizations. The system provided property management, tenant management, lease monitoring, and financial reporting functions. User evaluation indicated high satisfaction in terms of usability and operational effectiveness.

Likewise, Encarnacion, Villasin, and Encarnacion (2025) developed a unified web-based Property Management System designed to improve stakeholder services following the COVID-19 pandemic. The system integrated property registration, tenant management, maintenance requests, online payments, communication, document management, and reporting. The developed system achieved a high ISO/IEC 25010 evaluation rating and demonstrated the value of centralized property management platforms.

These studies demonstrate the growing importance of web-based property management systems in improving operational efficiency and customer service. Nevertheless, they focus primarily on rental property management and tenant services. The proposed study differs substantially because it supports the complete property acquisition process of a real estate developer—from reservation, documentary verification, billing, financing, and turnover preparation to association dues monitoring—while integrating OCR-assisted document validation and real-time inventory management.

Local Related Studies

OCR-Based Management Information Systems

Tumasis (2022) developed a web-based Management Information System integrated with Optical Character Recognition (OCR) for a Philippine accounting firm. The study aimed to automate the management of printed client records submitted to various government agencies. OCR technology reduced repetitive manual encoding and improved information retrieval by converting scanned documents into searchable electronic records.

The study supports the use of OCR for business process automation. However, it focuses primarily on information management within an accounting environment. In contrast, the proposed system applies OCR specifically to documentary requirements submitted during real estate transactions and combines OCR with document type validation and fuzzy string matching using the Levenshtein Distance Algorithm to improve validation accuracy.

Web-Based Property Management Systems

Caparos et al. (2025) developed a cloud-based Property Management System for real estate operations in the Philippines. The system centralized property records, tenant information, billing, and operational reports while improving organizational efficiency and tenant satisfaction. System evaluation showed positive results in usability, operational efficiency, and security.

Although similar in utilizing centralized property management, the developed system primarily addresses tenant management and property administration. The

proposed study differs by focusing on condominium sales transactions rather than rental operations. Furthermore, the proposed system introduces integrated reservation management, documentary validation, client portals, loan monitoring, association dues billing, and centralized document repositories specifically designed for St. Francis Square Realty Corporation.

Electronic Records Digitization

Pardiñan, Bondad, and Mangubat (2025) investigated the implementation of a web-based records digitization platform for Philippine State Universities and Colleges (SUCs). Their findings indicated that digitized document repositories significantly improved document retrieval, reduced dependence on paper-based records, and enhanced operational efficiency among institutional users.

Similarly, recent Philippine research on document tracking systems emphasizes that electronic document repositories improve organizational accountability, facilitate faster document retrieval, and minimize document loss by maintaining centralized electronic records.

These findings strongly support the centralized document repository implemented in the proposed study. Unlike conventional document tracking systems, however, the proposed system stores all reservation forms, Contracts to Sell, Deeds of Absolute Sale, financing documents, proof of payments, and Statements of Account within a centralized electronic client folder that is shared across multiple departments while remaining protected through Role-Based Access Control.

Comparative Analysis of Related Studies

Study	Major Contribution	Identified Limitation	Contribution of the Proposed Study
Lin et al. (2025)	OCR and automated document matching for real estate	Focused on document verification only	Integrates document validation with reservation, billing, client portal, and document management
Chinnasamy et al. (2025)	Secure OCR-based document management	Educational environment only	Applies OCR and RBAC to real estate operations
Sison et al. (2024)	OCR-enabled government document management	Government records only	Supports real estate documentary requirements and transaction workflow
Encarnacion et al. (2025)	Property management system	Rental and tenant management	Supports property acquisition, reservations, financing, billing, and turnover
Tumasis (2022)	OCR-based management information system	Accounting records only	Uses OCR with document type validation and Levenshtein-based validation in real estate transactions

Research Gap

The reviewed literature and studies demonstrate that recent information systems have successfully applied OCR, electronic document management, web-based portals, and property management technologies to improve organizational efficiency. However, no reviewed study was found to integrate all of the following capabilities into a single real estate management platform:

- A **Web-Based Real Estate Portal** and an **Internal Management System** sharing a centralized database.
- **Document Type Validation** before OCR processing to ensure that uploaded files correspond to the selected document category.
- **OCR integrated with the Levenshtein Distance Algorithm** to validate extracted information while accommodating minor OCR recognition differences.
- A centralized **Electronic Client Folder** that consolidates reservation forms, documentary requirements, contracts, financing documents, Statements of Account, and proof of payments.
- **Real-time property inventory synchronization** in which units automatically transition between **Available**, **On Hold**, **Sold**, and **Available** again after an approved cancellation while retaining the buyer's transaction history for audit purposes.

- Integration of reservation management, document management, billing, payment monitoring, loan monitoring, association dues monitoring, and report generation within a single system.
- A secure office-based Internal Management System supported by Role-Based Access Control and an internal company server, together with a Client Portal that provides controlled, view-only access to transaction-related documents.

These gaps justify the development of the proposed **OCR-Based Real Estate Management System for St. Francis Square Realty Corporation Using the Levenshtein Distance Algorithm for Automated Document Validation**. The proposed system combines document intelligence with comprehensive real estate business process automation, providing a unified solution tailored to the operational requirements of the company.

Synthesis

This is the summary of the related literature showing the gap and a need to develop or improve a system.

Guidelines for Writing the Synthesis of a Research:

--From all the literature and study you have reviewed you need to: Summarize key points from the literature and studies.

--Identify gaps, unresolved issues, or areas for improvement that your system aims to address.

--Justify the need for your study based on the findings from the reviewed materials.

Conceptual Framework of the Study

A conceptual framework illustrates the expected relationship between the variables (dependent and independent) in the research. It defines the relevant objectives for the research process and maps out how they come together to draw coherent conclusions. The conceptual framework should be constructed first before collecting data.

Provide a diagram/figure for this segment. Note that each figure/diagram should be followed by a discussion.

CHAPTER III

METHODOLOGY, RESULTS, AND DISCUSSION

Requirement Analysis

Requirements analysis is the initial stage in both the systems engineering and software development processes. It involves identifying and evaluating the needs, goals, and constraints of a new or modified system. This process takes into account potentially conflicting requirements from various stakeholders, including users, beneficiaries, and decision-makers.

In this section, provide a detailed discussion of the following feasibility areas, including the activities that will be conducted by the proponent to address each:

- **Operational Feasibility** – Assess how well the proposed system will function within the existing organizational environment.
- **Technical Feasibility** – Present the required software and hardware specifications for system development and implementation. These specifications should be organized in tables, followed by brief explanatory discussions for each.
- **Schedule Feasibility** – Include a Gantt chart to outline the project timeline and discuss how the schedule will support timely completion. The Gantt chart should be treated as a table and accompanied by a narrative explaining the key milestones and phases.

- **Economic Feasibility** – Analyze the projected costs and benefits of the system to determine its cost-effectiveness and sustainability.

Cost Benefit Analysis

This section estimates the cost of software, hardware, and labor required by the application of the project. It should compare the cost to implement such an application with the benefits delivered by the application. It should also demonstrate the values added to a given institution by the application of the project. Discuss the tangible and intangible benefits of the system.

Data Gathering Procedure

The development of the proposed **OCR-Based Real Estate Management System for St. Francis Square Realty Corporation Using the Levenshtein Distance Algorithm for Automated Document Validation** required a comprehensive understanding of the company's existing business processes, operational challenges, and system requirements. To obtain reliable and relevant information, the researchers employed four data gathering methods: **interview, observation, document analysis, and survey questionnaire**. These methods provided both qualitative and quantitative data necessary for analyzing the existing workflow and identifying the functional and non-functional requirements of the proposed system.

Interview

The primary source of information for this study was a series of informal and structured interviews conducted with selected personnel of **St. Francis Square Realty Corporation**. The interviews aimed to gather detailed information regarding the

company's existing business processes, departmental workflows, document handling procedures, reservation processing, billing operations, payment monitoring, and loan management activities.

The interviews also focused on identifying operational challenges encountered by employees, including repetitive manual encoding, multiple Excel-based monitoring files, physical document storage, delayed document retrieval, manual verification of documentary requirements, billing preparation, and coordination among different departments.

Information obtained during the interviews served as the primary basis for identifying the functional requirements of both the **Web-Based Real Estate Portal** and the **Internal Management System**.

A transcript of the interview and photographs or screenshots taken during the interview session are provided in **Appendix A**.

Observation

Direct observation was conducted to understand the actual workflow performed by different departments involved in the real estate transaction process. The researchers observed how reservations were processed, how documentary requirements were received and verified, how payment transactions were recorded, how Statements of Account were prepared, and how physical documents were stored and retrieved.

Observation also enabled the researchers to identify repetitive manual procedures, duplicate encoding of information among departments, document routing, inventory updating, and communication between various offices involved in the transaction process.

The results of the observation became the basis for designing the proposed workflow of the integrated system.

Photographs taken during the observation are presented in **Appendix B**.

Document Analysis

The researchers examined various company documents to obtain a comprehensive understanding of the information required for the proposed system. Existing forms, transaction records, reports, and operational documents were reviewed to identify the data fields, document formats, and business rules used by the company.

The documents analyzed included, but were not limited to:

- Reservation Application Forms
- Statement of Account
- Proof of Payment
- Contract to Sell
- Deed of Absolute Sale
- Official Receipts
- Notice of Acceptance
- Documentary Requirements Checklist
- Bank Financing Documents
- Pag-IBIG Fund Documents
- Billing Reports
- Collection Reports
- Association Dues Reports
- Existing monitoring spreadsheets

Document analysis assisted the researchers in designing the database structure, identifying required input fields, determining document validation requirements, and developing the electronic document management component of the proposed system.

Sample documents analyzed in this study are included in **Appendix C**.

Survey Questionnaire

A survey questionnaire was conducted to validate the need for the development of the proposed system and to determine the level of acceptance of the existing business processes among potential users. The survey gathered information regarding the efficiency of the current manual procedures, challenges encountered during daily operations, document management practices, reservation processing, billing, payment monitoring, and the perceived need for an integrated real estate management system.

The responses obtained from the survey were analyzed and used to support the justification for developing the proposed system.

A copy of the survey questionnaire is presented in **Appendix D**.

Survey Respondents

To validate the need for the development of the proposed OCR-Based Real Estate Management System, the researchers will conduct an online survey using **Google Forms**. The survey aims to gather the perceptions and experiences of the system's intended users regarding the company's existing processes and the proposed system.

The respondents will consist of **employees of St. Francis Square Realty Corporation** and **prospective property buyers**.

Employee Respondents

The employee respondents will be selected through **purposive sampling**, as they are directly involved in the business processes that will be automated by the proposed system.

A total of **twelve (12) employees** will participate in the survey, with **two (2) representatives from each department** that will utilize the proposed system.

Department	Number of Respondents
Sales Department	2
Loans Management Department	2
Documentation Department	2
Billing Department	2
Accounting Department	2
Cash Department	2
Total	12

These respondents are directly involved in reservation processing, client information management, billing, payment monitoring, document verification, loan processing, accounting, and cash management. Their responses will provide valuable insights into the operational challenges of the current manual processes and the potential benefits of implementing the proposed Internal Management System.

Prospective Buyer Respondents

The second group of respondents will consist of prospective property buyers who represent the intended users of the Web-Based Real Estate Portal.

Respondents will be selected using **convenience sampling** and must satisfy the following criteria:

- At least 21 years old;
- Interested in purchasing residential or commercial properties;
- Familiar with online transactions or web-based services;
- Willing to participate voluntarily in the survey.

The respondents will be grouped according to age:

Age Group

21–30 years old

31–40 years old

41–50 years old

Their responses will provide information regarding user expectations, online service preferences, convenience, accessibility, and acceptance of a web-based real estate portal.

The survey questionnaire will be administered using Google Forms, while the questionnaire and summary of responses will be included in **Appendix D** and **Appendix E**, respectively.

Discuss the various methods employed in gathering data for the system, clearly identifying the sources of each type of data. For interviews, ensure that a transcript and a photo or screenshot of the interview session are included in the Appendices. Additionally, reference the appropriate Appendix letter within this section for easy cross-referencing.

Provide a detailed profile of the individuals invited to participate in the survey for validating the need for system development. Include relevant demographic and

professional information that outlines their qualifications or experience related to the study. Ensure that a copy of the questionnaire used in the survey is included in the Appendices.

Analytical Tools

This part of the document should show the different diagrams used in the analysis and design of the system. The diagrams are as follows:

Data Flow Diagram. Start with the context diagram, DFD Level 0 followed by DFD Level 1. The context diagram should show the different entities (or users) involved in the system. Discuss first what DFD, in general, is all about. Each diagram must be followed by an explanation about the diagram itself. Apply the symbols by Gane and Sarson.

Use-Case Diagram. It is the representation of a user's interaction with the system that shows the relationship between the user and the different [use cases](#) in which the user is involved. Support the illustration with an explanation.

Activity Diagram. The activity diagram is another crucial behavioral diagram in the UML diagram to illustrate the system's variables. It is essentially a more complex flow chart that models the transition between activities.

Entity Relationship Diagram. This should show the database design of the system. Use the tabular form of ERD.

Algorithm

Discuss the algorithm to be used in the research. Provide a flow diagram on how the algorithm works.

System Architecture

Illustrate and discuss the conceptual model that defines the system's structure, behavior, and various perspectives. An architecture description provides a formal and organized representation of the system, enabling clear reasoning about its components and interactions. The corresponding illustration should be included in this section.

Development and Testing

Software Development. Specify the software development methodology adopted by the proponent (e.g., Scrum Methodology). Note that the use of the Waterfall Methodology is discouraged. Begin by discussing the key characteristics and principles of the selected methodology.

Then, provide a detailed explanation of the specific activities undertaken by the proponent at each stage or phase of the methodology. Ensure that each phase is clearly described in terms of its objectives, tasks performed, and deliverables produced. To enhance clarity, include a visual representation (figure) of the chosen software development methodology.

Software Evaluation. This section should discuss the various testing methods employed during the system's development, such as unit testing, integration testing, and user acceptance testing. Discuss the software evaluation tool used by the proponent.

Provide a detailed profile and number of respondents invited to evaluate the software, including demographic information, and relevant professional background that

supports their qualification to assess the system. The software evaluation form used in the study should be included in the Appendices; be sure to reference the corresponding Appendix letter within this section for easy cross-referencing.

Results and Discussion

This section should present the findings of the study, followed by a thorough analysis and interpretation of the data to explain the significance and relevance of the results. Offer a detailed discussion that highlights key insights, trends, and implications based on the evaluation outcomes. Include sample copies of the completed software evaluation forms in the Appendices, and ensure the corresponding Appendix letter is properly cited within this section for easy cross-referencing.

CHAPTER IV

CONCLUSION AND RECOMMENDATIONS

Briefly restate the main findings of your study, emphasizing the most significant results. Make sure the summary aligns with the objectives of the study, focusing on what was discovered or revealed.

Begin by reiterating the research problem or objectives. This reminds readers of the central focus of your study and sets the stage for summarizing your findings.

Briefly summarize the main findings of your research. Highlight the most significant results and how they contribute to the field. Avoid introducing new information; instead, synthesize what you have already presented.

Explain the broader implications of your findings. How do they advance knowledge in your field? What are the practical applications, policy implications, or theoretical contributions? Discuss how your research addresses the problem you identified at the beginning.

Conclude with a strong, impactful statement that emphasizes the importance of your research. This could be a thought-provoking remark, a call to action, or a suggestion for future initiatives based on your findings.

Be transparent about the limitations of your study. This shows academic honesty and helps contextualize your findings. It also provides a basis for future research.

Based on your findings, suggest areas for future research. What questions remain unanswered? What aspects of the problem could benefit from further exploration?

REFERENCES

This is a list of works cited, as well as works consulted but not cited (example, background reading not necessarily cited) in the construction of the research paper. The list of references is arranged alphabetically. For format on the writing of references, the APA (American Psychological Association) version 7th edition format of documentation shall be followed.

https://owl.purdue.edu/owl/research_and_citation/apa_style/apa_formatting_and_style_guide/reference_list_author_authors.html

Examples:

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APPENDICES

An appendix or appendices, if any, should be after the bibliography or references. Appendices include original data, preliminary tests, tabulations, tables that contain data of lesser importance, very lengthy quotations, forms and documents, computer printouts and other pertinent documents. Appendices should be arranged chronologically as they are cited in the main text. Use capital letters of the English alphabet to track appendices, and always begin with the letter A.

Appendix A
Transcript of the Interview

(Sample Template. Provide a photo of the interview.)

Appendix B

Background: Wendy Lee, manager of college services at Mountain View College, wants a new information system that will improve efficiency and customer service at the three college bookstores.

In this part of the case, Florence Fullerton (system analyst) and Harry Boston (student intern) are talking about justification for the new system and the project's feasibility.

Participants: Florence Fullerton and Harry Boston

Location: Mountain View College cafeteria, Monday afternoon, June 20, 2011

Project Status: Florence has received a systems request from Wendy Lee for a new bookstore information system.

Florence: Hi, Harry. Are you ready to get started?

Harry: *Sure. What's our next step?*

Florence: Well, when we analyze a specific systems request, we need to see how the proposal fits into the overall picture at the college. In other words, we have to analyze the business case for the request.

Harry: *What's a business case?*

Florence: A business case is the justification for a project. A strong business case means that a proposal will add substantial value to the organization and support our strategic plan.

Harry: *What's a strategic plan?*

Florence: A strategic plan is like a road map for the future. Without a long-range plan, it's hard to know if you're heading in the right direction. Our plan starts with a mission statement, which reflects our purpose, our vision, and our values.

Harry: *I see what you mean. I read the mission statement this morning. It says that we will strive to be an efficient, customer-friendly bookstore that uses a mix of interpersonal skills and technology to serve our students and support the overall objectives of the college. That says a lot in just one sentence.*

Manual Forms Used by the Company

Appendix C

Forms and Reports of the System

(These are the user interfaces of the system: forms are the input pages and reports are the output pages.)

Appendix D

Software Evaluation Forms

(Provide several (if not all) software evaluation forms accomplished by evaluators.)

Appendix E

Plagiarism Report

(This will be provided by the research adviser when the paper is accepted and approved by the panel. Only the first page of the plagiarism report will be included here.)

Appendix F
User's Manual

Appendix G
Curriculum Vitae